Campus Bakery Sales and Customer Preference Analysis

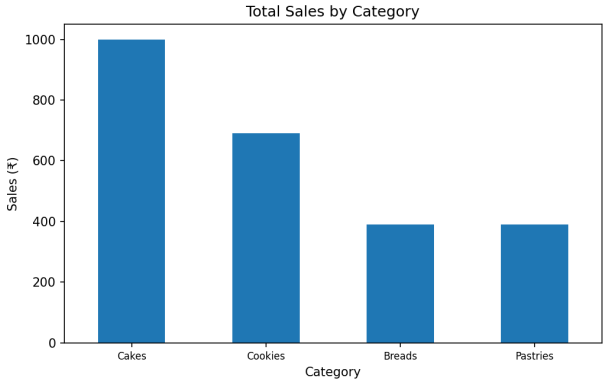
R Programming Tiny Project

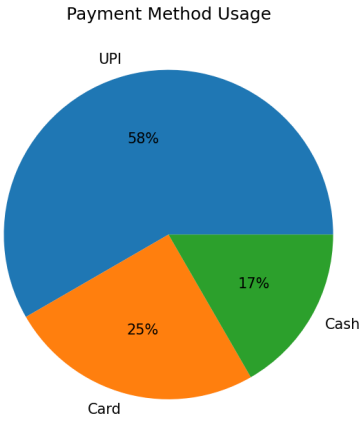
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Course	BCA (Hons.)
Subject	R Programming
Project Type	Tiny Project – R Markdown
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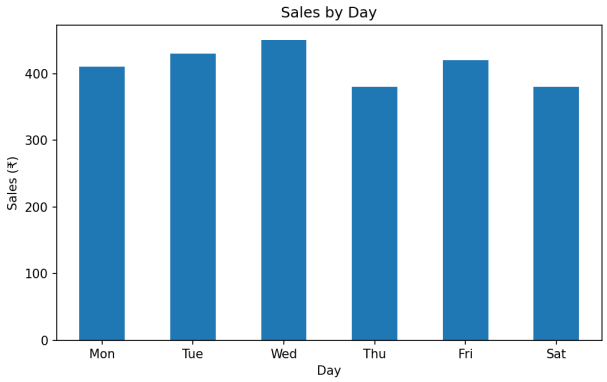
This report uses a synthetic campus bakery dataset to demonstrate an end-to-end R programming workflow.

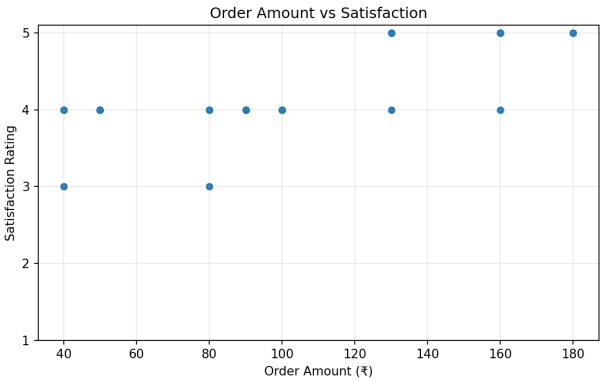
Project Dashboard

24 Orders	₹2,470 Total Sales	₹103 Avg Order	4.21/5 Avg Rating









1. Introduction
This tiny project analyses a synthetic campus bakery order dataset using R. It demonstrates data importing, preprocessing, statistical exploration, visualisation and interpretation.

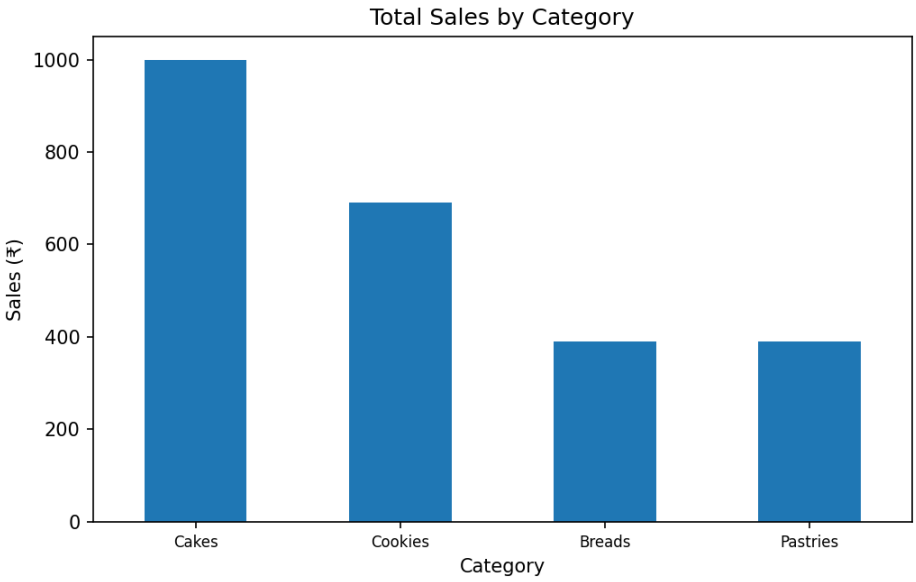
2. Objectives
Import and inspect data; clean missing values; calculate descriptive statistics; analyse sales and payment patterns; create graphs; interpret findings.

3. Dataset
The dataset contains 24 orders and 7 variables: Order ID, Day, Category, Product, Payment Method, Amount and Satisfaction Rating.

4. Data Preprocessing and Cleaning
Missing values were checked and incomplete records removed. Categorical fields were converted to factors.

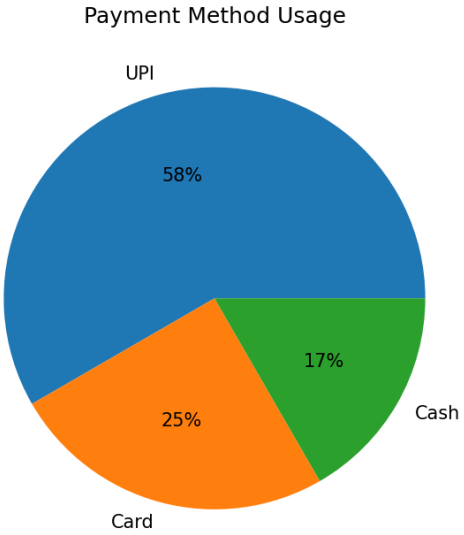
5. Statistical Analysis
Total sales: ₹2,470. Mean order: ₹102.92. Median: ₹95.00. Standard deviation: ₹46.39. Average satisfaction: 4.21/5. Correlation between amount and satisfaction: 0.74 (moderate positive). The correlation is descriptive and does not establish causation.

6. Sales by Category
Cakes has the highest total category sales (₹1,000).



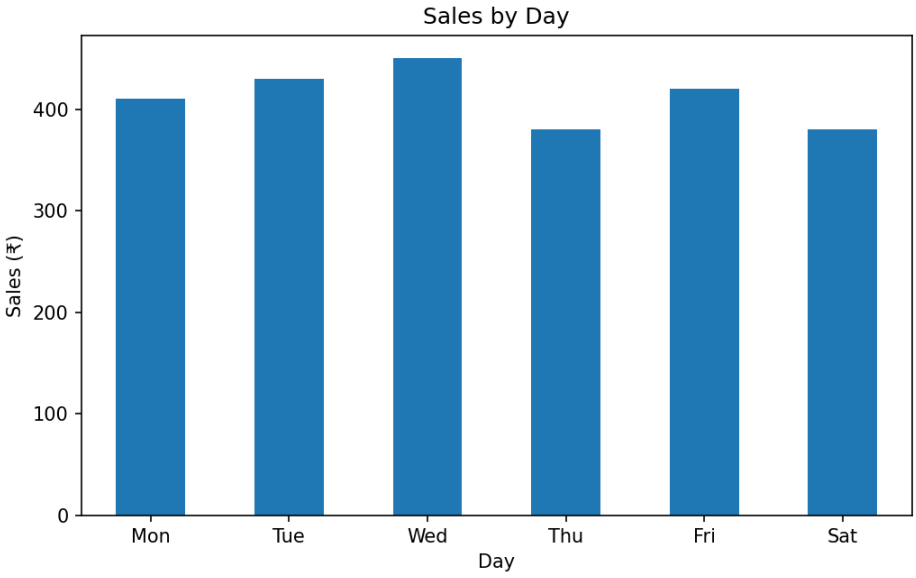
7. Payment Method Analysis

UPI is the most frequently used payment method in the sample (14 of 24 orders).



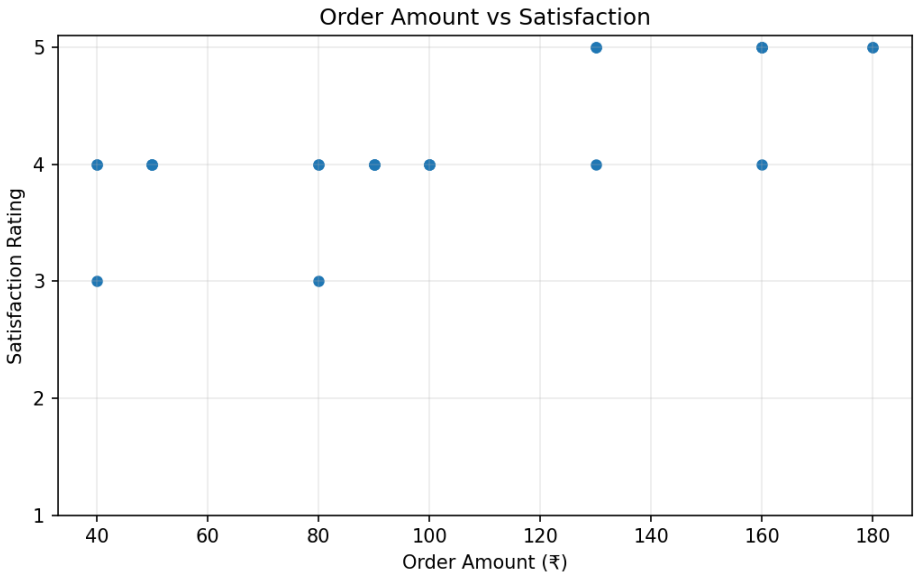
8. Daily Sales Analysis

The highest sales day is Wed with ₹450.



9. Order Amount vs Satisfaction

The observed correlation is 0.74. The small synthetic sample means this is descriptive only.



10. Findings

- Total recorded sales are ₹2,470.
- The average order value is ₹102.92.
- Cakes has the highest category sales.
- UPI is the most common payment method.
- Chocolate Slice has the highest product-level sales.
- Average satisfaction is 4.21/5.

11. Conclusion

The project demonstrates how R can import, clean, analyse and visualise structured sales data. Descriptive statistics and graphs provide a clear summary of sales patterns and customer preferences.

12. R Functions Used

Function	Purpose
read.csv()	Import CSV data
head()	View first rows
str()	Inspect structure
summary()	Statistical summary
is.na()	Check missing values
complete.cases()	Remove incomplete records
mean()/median()	Central tendency
sd()	Standard deviation
cor()	Correlation
aggregate()	Grouped summaries
barplot()	Bar charts
pie()	Pie chart
table()	Frequency counts
plot()	Scatter plot